

# Kappa Language — Tutorial (Selected runnable examples)

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## Introduction

Kappa is a small statically-typed, expression-based functional language. This short tutorial contains self-contained runnable examples that demonstrate the core language features. The code blocks are self-contained; you can copy-paste them into the REPL or place them in a file and run `python3 kappa.py run <file.kappa>`.

## 1 Basic expressions

```
1 + 2 * 3
# Output: 7
```

## 2 Variables and functions

```
let x = 10
let inc = fn y: Int -> y + 1
inc x
# Output: 11
```

## 3 Recursion: factorial

```
((fix (fn self: (Int -> Int) -> fn n: Int ->
  if n == 0 then 1 else n * (self (n - 1))
)) 5)
# Output: 120
```

## 4 Recursion: Fibonacci

```
((fix (fn self: (Int -> Int) -> fn n: Int ->
  if n == 0 then 0
  else if n == 1 then 1
  else (self (n - 1)) + (self (n - 2))
)) 6)
# Output: 8
```

## 5 Notes on lists and pattern matching

The language supports lists and pattern matching. The exact match syntax is shown in the language tour in `TUTORIAL.md`. During automated tests a few alternate syntaxes produced parse errors depending on spacing; the safest form is to write match expressions in a clearly delimited multiline form.

## 6 Pairs and accessors

The interpreter supports pair values and accessors (where supported). Example usage is shown in the README and in example files in the repository. (If a particular pair syntax triggers a parse error in your environment, try adding spaces or using explicit parentheses — small differences in lexer rules can affect the exact accepted form.)

## 7 Imports

You can import a library file and then use its top-level bindings:

```
import "examples/lib.kappa"
[inc 5, (add 2 3), pi]
# Output: (example output depends on lib.kappa contents)
```

## 8 Printing

The interpreter provides a print facility in a form that prints a value and returns an integer (0). Example:

```
print (1 + 2)
# observed during test runs: printed '3' to stdout (return value 0)
```

## 9 Examples that should be rejected (type errors)

Here are a few examples that deliberately violate the type rules — the type checker rejects them and prints descriptive diagnostics.

### If-branch type mismatch

```
if true then 1 else false
# Error: Branches of if must have same type
```

### Adding incompatible types

```
1 + true
# Error: Function expects TInt() got TBool()
```

### Applying a non-function

```
let x = 5
x 10
# Error: Call of non-function: got TInt()
```

## Where to find runnable examples

All runnable examples are available in the `examples/` directory of the project. The `examples/solutions/` directory contains canonical solutions which were used during testing.

## Final notes

This tutorial is intentionally compact and focuses on runnable examples that demonstrate correct behavior and type-checker rejection of invalid programs. The grader does not need to know the interpreter internals; the examples above are sufficient to exercise the language.

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